Review paper

Plant disease detection is a key area of research because crop diseases can lead to significant losses for farmers and impact food production. Traditionally, farmers relied on experts to examine plant leaves and identify diseases. This process was time-consuming and needed specialized knowledge. To address this issue, researchers are now using Artificial Intelligence, particularly Convolutional Neural Networks (CNNs), to automatically detect plant diseases from leaf images. In these systems, a farmer can take a photo of a leaf, and the AI model will analyze it to identify the disease. Techniques like transfer learning and data augmentation help improve accuracy and manage variations in lighting, angles, and image quality. Many studies have tested these models on crops such as rice, potato, tomato, and various tropical plants, with most showing promising results. These AI-based systems are being integrated into mobile applications, allowing farmers to upload leaf images and receive immediate disease detection along with treatment suggestions. This makes the technology practical and accessible, especially in rural areas where agricultural experts may not be available. However, challenges remain. Real-world conditions such as poor lighting, shadows, and complex backgrounds can affect accuracy, and pinpointing the exact stage of a disease is still difficult. Despite these challenges, AI-powered plant disease detection systems provide a fast, reliable, and cost-effective way to reduce crop loss, improve yield, and support the advancement of modern agriculture.

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